

Machine Learning-Based Predictive Analytics for Early Detection of Diabetes

ABSTRACT

Diabetes is a major global health challenge affecting millions of individuals worldwide. According to the World Health Organization (WHO), approximately **830 million people are living with diabetes globally**, and the prevalence among adults has increased from **4.7% in 1980 to nearly 8.5% in recent years**. Diabetes is responsible for nearly **1.5 million deaths annually** and is a leading cause of serious complications such as heart disease, kidney failure, and stroke. The rapid increase in diabetes cases highlights the need for early detection and preventive healthcare solutions.

This project proposes a **Machine Learning-Based Diabetes Prediction System** designed to predict the likelihood of diabetes using patient health parameters such as glucose level, blood pressure, body mass index (BMI), insulin level, and age. The system applies supervised machine learning algorithms to analyze medical data and identify hidden patterns associated with diabetes risk. The proposed system aims to assist healthcare professionals and individuals by providing an efficient early risk assessment tool. By integrating machine learning techniques into healthcare diagnostics, the project contributes to improving early detection, reducing complications, and supporting preventive healthcare strategies.

Keywords: Diabetes, Predictive Healthcare Analytics, Machine Learning, Early Risk Detection.

TEAM MEMBERS

ANBU SELVAN M (24CSR021)

BHARANIDHARAN D (24CSR037)

BHAVANA R (24CSR040)

MENTOR NAME

Dr.N.SHANTHI